

Data Management Plan

1. Organizational context

Name	Project Group 08 – NeuroPark Company
Group members	Anthony Jansen, Iris Lagemaat, Pascal Dubbelman, Polly Cheung, and Susanna van de Wetering
Date	25-06-2026
Chair group	Geo-Information Science and Remote Sensing (GRS)
GRS Supervisor	Erick Chacón-Montalván, Harm Bartholomeus
Start date project	01-06-2026
File name DMP	NPCa2_DataManagementPlan

2. Project description

Title	The Increasing Pressure on Parking Spots - <i>Mapping Parking Spaces in Wageningen using Geoinformation and AI</i>
Description	The municipality of Wageningen aims to reduce parking pressure. To achieve this, they need an accurate and up-to-date overview of parking inventory, especially the physical supply and location of the individual parking spaces. However, creating and maintaining such an inventory through traditional methods is time-consuming and difficult to scale. With the rapid development of urban remote sensing and deep learning–based methods, their efficiency and effectiveness now enable support for mapping parking inventories. However, challenges remain regarding data inputs and detection accuracy. Many existing deep learning studies focus on vehicle-mounted or surveillance camera imagery for parking space detection. Furthermore, current research mostly focused on whole parking lots, while the municipality is interested in individual parking spaces. This four-week-project delivered a proof-of-concept on using deep learning and aerial imagery for mapping individual parking spaces in parking lots and marked on-street parking in Wageningen. The project used open-source aerial imagery from Beeldmateriaal Nederland. The images were annotated to create a small training and validation dataset. Additionally, the models were tested on an unseen dataset and finally used for prediction of the whole Nude neighbourhood. Different deep learning architectures were implemented to find the most suitable workflow for this project context. The

	workflow is documented and presented to the municipality, so that the methodology can be repeated and expanded on in the future. The output should be interpreted as a draft mapping workflow that can support municipal inventory work, instead of a fully automatic replacement for manual verification.
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3. Data management roles

Data collecting	Anthony Jansen, Iris Lagemaat, Pascal Dubbelman, Polly Cheung, and Susanna van de Wetering
Data analysing	Anthony Jansen, Iris Lagemaat, Pascal Dubbelman, Polly Cheung, and Susanna van de Wetering
Other contributions	No
Role of supervisor	Supervision of the ACT project, suggestions, no data collection/analysis/storage

4. Project data

Data stage	Specification of type of project data	Software choice	Data size/ growth estimation
Source data	Multiple 1x1km georeferenced aerial images (resolution: 7,5 cm). (https://www.beeldmateriaal.nl/)	Image Viewer/QGIS (.tif)	140 MB per image
	Georeferenced parking polygons available on Wageningen Municipality website (Parkeervlakken - Overzicht)	QGIS/ArcGIS (.geojson)	3 MB
	ESRI Standard basemap	QGIS	500 MB
Collected Data	Processed tiles selected for annotation	ArcGIS (.tif/ .tfw)	650 MB
	Manual annotations of bounding boxes corresponding to processed tiles	ArcGIS (Pascal VOC XML bounding boxes)	1 MB

Result data	Predicted parking-space bounding boxes from YOLOv9c and YOLOv11	QGIS (.geojson)	1 MB
	Accuracy metrics (precision, recall, F1-score, Average Precision, mAP@50, mAP@50-95) to evaluate model performance for YOLOv9, YOLOv11s HBB, YOLOv11s OBB	.csv	30 kB
	Final maps, figures, presentation, report	PDF, images	Under 1 GB
Models/ Code	Preprocessing script to split image into tiles	Jupyter notebook	10 kB
	Preprocessing script to rename the tiles after the annotation phase	Jupyter notebook	19 kB
	Preprocessing script to randomly select test images in the testing phase	Jupyter notebook	6 kB
	Preprocessing script to combine the GeoJSON produced in the annotation phase	Python script	5 kB
	Script for training, validating and testing YOLOv9c	Jupyter notebook	8 kB
	Script for training, validating and testing YOLOv11s with HBB	Jupyter notebook	8 kB
	Python utility scripts needed for the YOLOv9c and YOLOv11s with HBB Jupyter notebooks for image conversion from GeoTIFF to PNG, annotation conversion, train/validation splitting, model training, and producing georeferencing predicted bounding boxes	Python scripts	34 kB
	Script for training, validating and testing YOLOv11s with OBB	Jupyter notebook	36 kB
	Script for exploratory post-processing for analysis of	Jupyter notebook	208 kB

	duplicate detection bounding boxes for YOLOv11s OBB		
	Python utility scripts needed for the YOLOv11s with OBB Jupyter notebook for image conversion from GeoTIFF to PNG, annotation conversion from GeoJSON to YOLO-OBB label format, train/validation splitting, model training, prediction and conversion of OBB predictions back to GeoJSON	Python script	60 kB

5. Short term storage

Data stage	Storage location	Backup procedures
Source data	Teams (OneDrive)	External SSD/USB
Result data	Teams (OneDrive)	External SSD/USB
Models/code	GitHub (ParkingSpaceDetection)	Teams (OneDrive)
Other	Teams (OneDrive)	External SSD/USB

6. Data Naming Structure

The workspace is divided into different folders where the following naming structure is used:

NPC (NeuroPark Company) + ModuleLetter + Number + Name

All data will be stored in a ZIP-file, except the scripts, which can be found on the Git Repository. The following tables display the folder structures.

ZIP-File	
NPCa_Documentation	NPCa0_Readme.md NPCa1_MetaData_ISO19115.xlsx NPCa2_DataManagementPlan.pdf
NPCb_SourceData	NPCb0_GISlayers NPCb1_AerialImagery
NPCc_CollectedData	NPCc0_Images NPCc1_Labels NPCc2_PrintedMaps
NPCd_ModelsScripts	NPCd: see Git Repository
NPCe_WorkEnviroments	NPCe0_QGIS (containing the raw predictions NPCe1_Flowcharts (<i>containing the literature review flowchart</i>)
NPCf_IntermediateResults	NPCf0_GeoJSONs (<i>containing the predictions as GeoJSON-files</i>) NPCf1_JPGs (<i>containing the predictions maps as JPG-files</i>)
NPCg_Reports	NPCg0_Proposal.pdf NPCg1_DraftReport.pdf NPCg2_FinalReport.pdf NPCg3_FinalPresentation.pptx

Git Repository	
Create_Tiles	NPCd0_SplitTiles.ipynb NPCd1_RenameTiles.ipynb NPCd2_RandomSplitTestImages.ipynb
Combine_Geojsons	NPCd3_CombineGeojson.py
Object_Detection_Models	NPCd4_YoloV9_Pipeline.ipynb NPCd5_YoloV11_Pipeline.ipynb NPCd6_YoloV11_OBB_Pipeline.ipynb PythonHBB (<i>containing functions as PY-files</i>) PythonOBB (<i>containing functions as PY-files</i>)
Explo	NPCd7_YoloV11_OBB_Nude_PostprocessExperiments.ipynb
Reports	NPCa2_DataManagementPlan.pdf NPCg2_FinalReport.pdf NPCg3_FinalPresentation.pdf
LICENSE	
README	

7. Documentation and metadata

The source data metadata is documented in Excel sheet found in NPCa1_Metadata_ISO19115. The team will adapt this sheet so that it captures metadata fields according to the ISO 19115 standards.

The metadata elements include:

- Identification
- Extent
- Quality
- Spatial Reference
- Content
- Distribution

The resulting data files will comprise aerial imagery tiles, manual parking-space annotations, converted YOLO label files, prediction outputs in forms of bounding boxes, georeferenced vector layers, maps, python notebooks of implementing pipelines using YOLOv9c, YOLOv11s with HBB, YOLOv11s with OBB, georeferenced vector layers, Python utility scripts, and processed results. Trained model weights are generated as intermediate outputs but are not stored in the managed project data storage.

8. Sharing and Ownership

Sharing and ownership	With who/what/how
Data sharing	Results (outputs) are shared with the Municipality of Wageningen. These results include Git repository, the annotated datasets, the output GeoJSON, the report, and the final presentation.
Data ownership	Source data belongs to their original copyright holder. The collected data, i.e. the annotations, processed training data and project results will be in ownership of the Municipality of Wageningen.

9. Long term storage

Yes or no?	Argumentation
Yes	Final project data will be delivered to the Municipality of Wageningen; The Git repository is publicly available. Long-term storage by the Municipality is required to ensure the workflow can be reproduced, verified, and extended in future work, and to enable the municipality to further develop and build upon the workflow in the future.